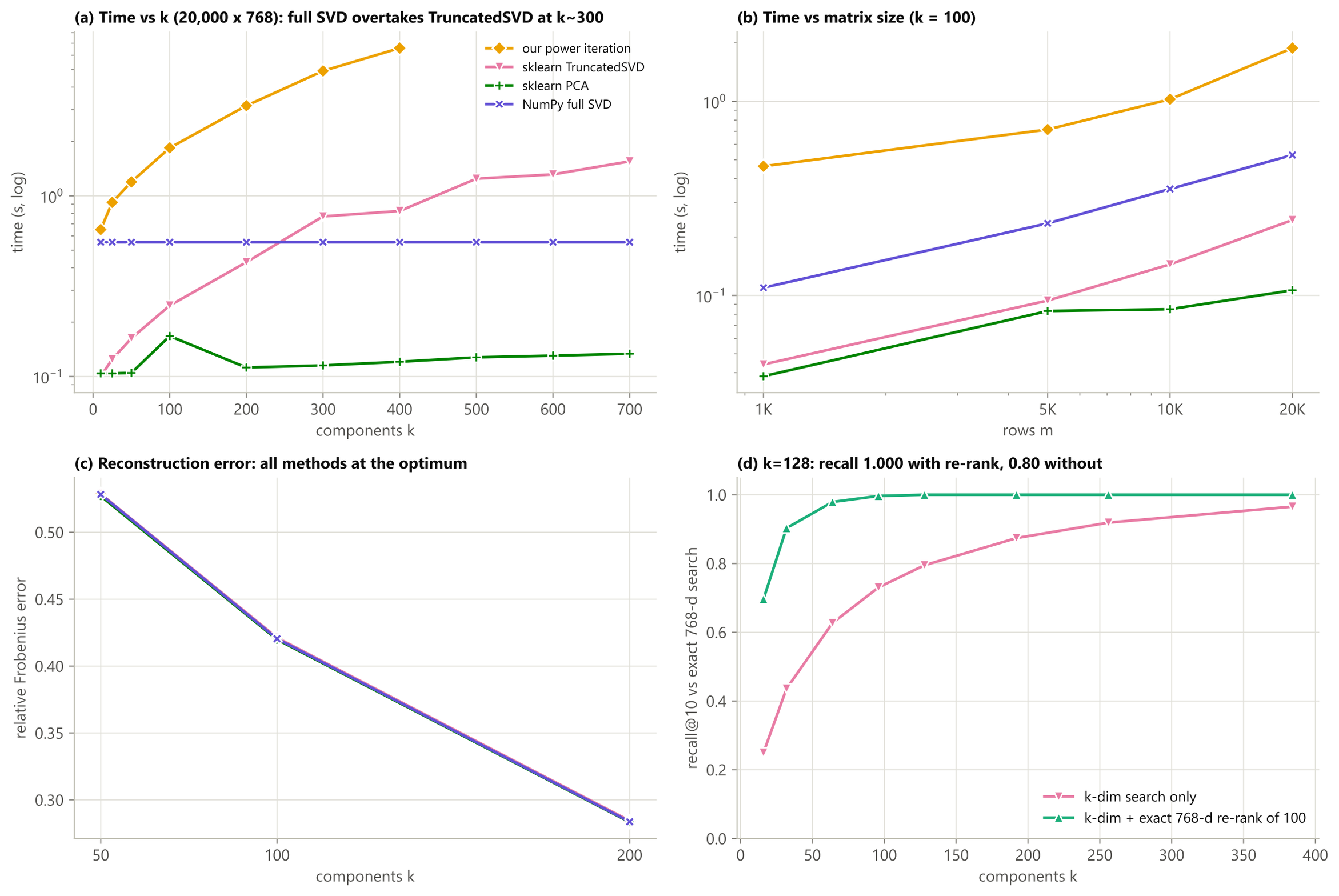
(a) Time vs k (20,000 x 768): full SVD overtakes TruncatedSVD at k~300
our power iteration
sklearn TruncatedSVD
sklearn PCA
NumPy full SVD
time (s, log)
components k
(b) Time vs matrix size (k = 100)
time (s, log)
rows m
(c) Reconstruction error: all methods at the optimum
relative Frobenius error
components k
(d) k=128: recall 1.000 with re-rank, 0.80 without
recall@10 vs exact 768-d search
k-dim search only
k-dim + exact 768-d re-rank of 100
components k